Mathematical overview: taking turns is not tossing a coin

One switch alternates. Follow the visible arrow, then flip that card; do not reset it between counters. Every pair of consecutive departures sends one counter each way. The exit piles are therefore equal or differ by one, and two consecutive counters never share an exit. Children can explain this by showing the changed arrow and pairing arrivals, without using odd/even vocabulary.

The four-exit tree repeats after four counters. Its top card sends two counters to each lower card; each lower card sends one to each exit. Every card flips an even number of times and returns to its start. Thus successive blocks of four reach A, B, C and D once, in the same order; eight give two each and twelve give three each. This needs the printed equal-depth tree, one counter finishing before the next starts, and no manual card changes within a trial. It is not a claim about arbitrary trees or roads.

The start controls order, not just totals. The three cards have eight possible starts, giving eight different four-arrival orders. The first two counters always enter opposite halves. A route to one chosen exit fixes only the two cards it touches; the unused lower card is free. A coin, by contrast, can choose either exit anew: repeated exits and unequal piles are possible. Fair, independent tosses are assumed only for the probability notes below.

Destination and evidence. K–1 finds that the switch never repeats an exit: its piles differ by at most one, the extra counter follows the first arrow, and after counter 1 every later exit is predictable. Four tree counters always reach four exits; two chosen cards direct the first counter. The older tables' larger trees and road score (base guide p. 1) are not prerequisites. Trials are experiments; "it always happens" is a conjecture; forced turns and paired arrivals explain it. Successful trials prove possibility. Failure proves impossibility when every start has been tried: both switch arrows, or all eight tree starts pooled across pairs. Tosses that never show a result prove nothing.

Preparation and adult support

Status. Unpiloted. Actual-size printing, cutting/folding/gluing, card flipping, spare-card records, counter clearance and exit piles, staffing and timings are physically unrehearsed.

Per pair. Print one copy of `k-1.pdf` (seven pages). From `week-91-materials.pdf` (F91-M-v1), print pp. 1, 2 and 6: the one-switch board, landscape four-exit board and seven tree cards. Fold/glue only the seven tree pairs into 18×22 mm reversible cards; keep NEXT upright and flip side-to-side. Three cards operate the tree; four remain spare, of which three preserve its start. Add 16 counters no wider than 12 mm, a coin and pencil. Optionally supply two arrival strips per pair from p. 7 (twelve 15 mm squares each), for Problems 4 and 7. Print single-sided at 100%, retaining landscape orientation. Student pictures are records, not card mats: the kit has 32 mm switch/exit circles on p. 1, and 24 mm switches with 30 mm exits on p. 2.

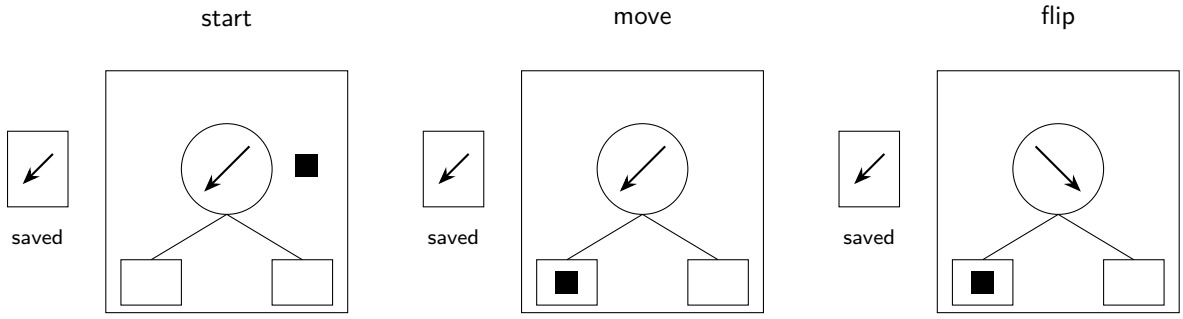
Two K–1 kits, additional to older tables. Prepare 14 tree cards, 32 counters, two coins, two copies each of materials pp. 1, 2, 6, and one copy of p. 7 (four strips). Allow about 30 minutes beyond the week's existing preparation; this is untimed. Use scissors and glue to make cards; bring scrap paper for the demonstration. Roads and the eight-exit board stay off this table.

Before use, measure p. 7's 50 mm bar and rehearse flipping with a waiting counter beside each card. If exit piles crowd the circles, keep them beside their matching letter or in labeled cups, as the base guide allows.

Prerequisites and staffing. Count to 12, match named exit letters, follow an arrow and retain "move, then flip." No independent reading is needed. One adult reads and, when needed, writes children's reported arrivals or arrows; children choose starts, move and flip, with a partner checking. Rotate roles. With three children, use a pair plus a rotating checker; keep the second kit ready. If two pairs both need a scribe, stagger recording tasks or work as one team rather than let the adult run either machine.

Launch, a first visit and the one-switch key

Join the shared launch. After the usual movement opening, use the base guide's 3–5 minute whole-group demonstration: handle a counter and spare card, follow the arrow, then flip, keeping the counter beside the card. At the K–1 table, spend two setup minutes on a scrap-paper switch with blank exits and a circle at least 24 mm across, as on student p. 1. Show just one departure, not a numbered task or a ten-counter answer. Copy its starting arrow with a spare beside the board; only the working card flips.



The black square is the counter. The saved arrow does not change. Once exits X and Y are named, one arrival fills one box; the adult may write the child's letter. On the tree, keep the three saved cards in a matching top/left/right arrangement *off* the board. Copy from these, never from changed working cards. Keep later arrivals waiting while recording catches up.

A one-hour first route. Allow 5 minutes each for movement, shared launch and table setup (including the two-minute demonstration), then 20–25 for Problems 1–2 and 5–10 for Problem 3. Reserve 5 for sharing and packing. Start Problem 2 with pictures 1, 3 and 4: one for each arrow and one impossible picture. Try Problem 5 only if 15 minutes remain before packing and partners manage flips; otherwise return to it next time. Stay with the switch if needed. Problem 4 and Problems 6–9 are further visits, not a first-hour quota.

Keep records light. The full pages offer up to 40 letters in P1, 42 in P3, 16 prediction entries and 78 arrow circles. One written row from each arrow is enough. Watch two-in-a-row and read P3's five-and-two answer from exit piles without writing. For a slanted arrow, accept a slanted stroke or pointing to a saved card while the adult draws.

Problem 1, student page 1: ten arrivals

From first arrow X: `XYXYXYXYXY`. From Y: `YXYXYXYXYX`. Neither has two consecutive arrivals at the same exit; each leaves five at each exit. A child may point to the flipped arrow as the reason the next arrival must change. Ask only if needed: "Where is the arrow pointing now?" or "Could that same exit be next without changing a rule?" Later ask whether any exit can ever be two ahead.

Problem 2, student page 2: all six pile pictures

Read pictures left to right, top row then bottom row. X is left, Y is right.

Picture	1	2	3	4	5	6
Piles X,Y	2,1	3,3	2,3	5,2	5,4	3,5
Start	X	either	Y	impossible	X	impossible

The two starts exhaust the choices. Pair arrivals: each pair supplies one to each exit, with at most one unpaired counter going to the first arrow's exit. This rules out pictures 4 and 6. Use the saved card when drawing the start: after an odd number of counters the working card points the other way. Hint: "Can you match each X counter with a Y counter? How many are left unmatched?"

Coins, predictions and the four-exit tree

Problem 3, student page 3: a coin can repeat

Both answers are yes. Agree aloud that heads means X and tails Y, then put the switch card away. For example, the possible toss order heads, heads, heads, heads, heads, tails, tails makes `XXXXXYY`: two consecutive counters share an exit, and the piles are five and two. A child may invent any other witness.

There are six seven-box rows, not a requirement to keep tossing until a success. For fair independent tosses, 42 of the 128 seven-toss orders give five and two in either direction. Even six trials miss that split about 9.2% of the time. If it never appears, ask "What faces would make the piles? Could the coin show those faces?" A short run cannot establish impossibility or that a coin must balance. Keep the counts as adult background; children need only the possible face-by-face story.

Problem 4, student page 3: predict before continuing

The printed table lists counters 5, 8, 7 and 4 in that order. After watching counter 1, make *all four guesses first*. Continue the same trial through counter 8, without resets; the adult fills "landed" when counter 4, then 5, then 7, then 8 arrives. Thus the out-of-order rows do not require going backwards or separate trials. Keep a single arrival strip if needed.

Counter	5	8	7	4
Card, compared with counter 1	same	other	same	other

If counter 1 was X, the four predictions are X, Y, X, Y; if Y, they are Y, X, Y, X. The arrow alternates at every departure, so returning in pairs explains the pattern. For the coin there is no guaranteed prediction: either exit is possible at each listed counter, regardless of the first toss. Under the fair independent model each guess has probability one half, not certainty. Hint: "What does the card remember that the coin does not?" Correct guesses alone do not make the coin predictable.

Problem 5, student page 4: can two of four share an exit?

Before this task, a child sends one counter while the adult names both stops: wait beside the top card, move, flip it; wait beside the lower card, move, flip that card. If a wrong card is flipped or reset during a trial, clear exits, restore the saved start and begin again.

No, from any start. Two go to each lower card, and each lower card sends its two to different exits. Let children show that with four counters before asking why. Ask "How many reached each lower card? What did that card do with its two?"

For adult lookup, L means down-left and R down-right. A three-letter word lists **top, left lower, right lower** cards, not exit letters. Use physical saved cards with children instead. The complete list is:

Start	First four arrivals	Start	First four arrivals
`LLL`	`ACBD`	`RLL`	`CADB`
`LLR`	`ADBC`	`RLR`	`DACB`
`LRL`	`BCAD`	`RRL`	`CBDA`
`LRR`	`BDAC`	`RRR`	`DBCA`

Each of three cards has two choices, giving $2 \times 2 \times 2 = 8$ starts. The student page has nine recording trees: one is spare, not evidence for a ninth start. If the children want a complete check, pool distinct starts across pairs and use this page's saved drawings again for Problem 6; do not require every child to repeat all eight. Four counters also return every card to its start, so counters 5–8 repeat the order. A natural continuation is to predict the next four before sending them.

Pile targets and programming the start

All card words below use top, left lower, right lower order, as defined on guide p. 3. A list gives *every* successful start, though most problems ask for just one.

Problem 6, student page 5: six pile pictures

Read each row left to right, then move down. Piles are in A, B, C, D order.

Picture	Counters	Piles	All successful starts
1 (top left)	8	2,2,2,2	all eight
2 (top right)	6	2,1,2,1	`LLL`, `RLL`
3 (middle left)	6	2,2,1,1	impossible
4 (middle right)	12	3,3,3,3	all eight
5 (bottom left)	6	1,2,1,2	`LRR`, `RRR`
6 (bottom right)	8	3,1,2,2	impossible

Four-counter blocks give one to each exit, so pictures 1 and 4 always work and picture 6 never does. After six, the top card has sent three to each side; picture 3 wants four left and two right. For picture 2 the lower cards must first favor A and C; for picture 5, B and D; either top direction works. Hint: "How many counters would this picture need on each side of the top card?" Have children copy saved arrows, not the lower cards that have just flipped three times.

Problem 7, student page 6: B first or last to get two

B is first to two at counter 5 from `LRL` or `LRR`; their orders begin `BCAD` or `BDAC`. B is last to two at counter 8 from `RLL` or `RLR`; their orders begin `CADB` or `DACB`. Counter 5 lands where counter 1 did, and no exit gets a second before all four have one. The first exit to reach two is counter 1's; the last is counter 4's (at counter 8). B first needs B as counter 1; B last needs B as counter 4. Hint: "When the next round starts, which exit will be visited first?" The start list proves these are all the choices. Draw from the saved cards before testing, even if stopping as soon as B gets two.

Problem 8, student page 6: every start with C first

Exactly two complete settings: `RLL` and `RRL`. The top card must send the counter right; the right lower card must send it left to C. The left lower card is never touched, so either direction works. "The left card doesn't matter" is the structural answer; accept that insight, then ask the child to draw both settings. The third printed tree is spare space, not a missing third answer. Hint: "Which card did your counter never visit?" Once ready, children can choose a different first exit and ask which card becomes free.

Problem 9, student page 7: the first two exits

Top left, A then D: `LLR` only. Top right, D then C: impossible. Bottom left, C then B: `RRL` only. Bottom right, B then A: impossible. The first two counters must go to opposite sides of the top card; C and D share one side, as do A and B. In a possible picture, the first counter fixes the top and one lower card; the second fixes the other lower card, so no choice is left. Hint: "Where must counter 2 go when the top card has flipped?"

Record first use. Note preparation time, problems reached, missed or wrong flips, accuracy of saved-card copies, who wrote, and which pictures children crossed out and why.

Where it goes. Once the pages are approved, this section replaces "K–1 route, adult-read" under "Preparation, launch and visit choices" on page 2 of `week-91-facilitator.pdf` (F91-FAC-v3), including its final two lines at the top of page 3. Also update p. 1's K–1 overview sentence to point to this section instead of saying "not an independent packet." Keep the shared launch and older-band solutions.